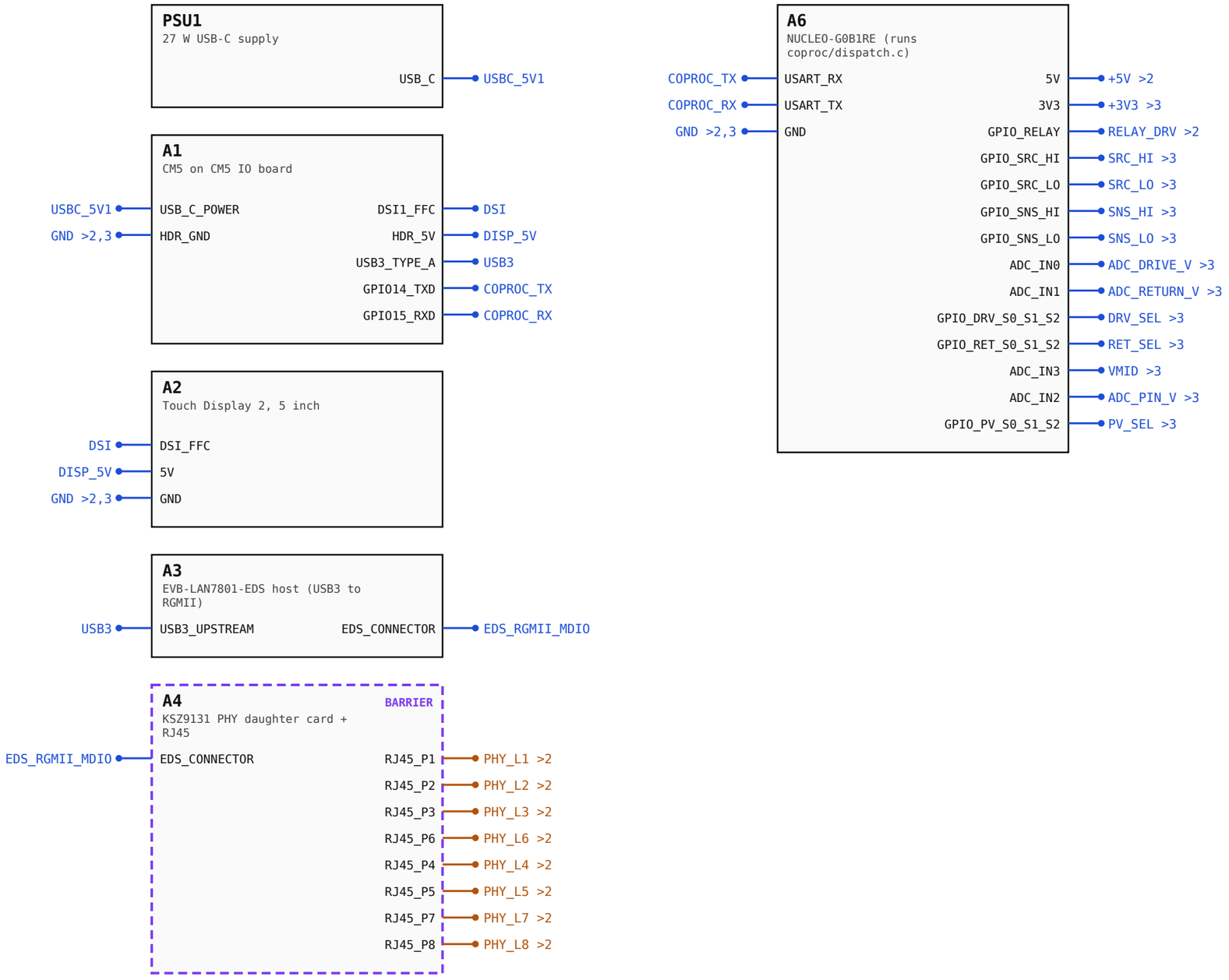


NetScope X1 bench kit - wiring schematic

Processor, display, test-port MAC and PHY

Not built. Confirm every eval-board connector against its user guide before power.

SYS system side · LINE cable conductors · POE network-powered

Block level: function-named pins, not package pins

J1

RJ45 breakout: patch lead to A4

PHY_L1 >1	P1
PHY_L2 >1	P2
PHY_L3 >1	P3
PHY_L6 >1	P6
PHY_L4 >1	P4
PHY_L5 >1	P5
PHY_L7 >1	P7
PHY_L8 >1	P8

K1

BARRIER

DPDT signal relay 5 V coil, pair 1-2

+5V >1	COIL_PLUS	COM_A	CUT_1
PHY_L1 >1	NC_A	COM_B	CUT_2
PHY_L2 >1	NC_B	NO_A	T1 >3
		NO_B	T2 >3
		COIL_MINUS	RELAY_COIL_LOW

K2

BARRIER

DPDT signal relay 5 V coil, pair 3-6

+5V >1	COIL_PLUS	COM_A	CUT_3
PHY_L3 >1	NC_A	COM_B	CUT_6
PHY_L6 >1	NC_B	NO_A	T3 >3
RELAY_COIL_LOW	COIL_MINUS	NO_B	T6 >3

K3

BARRIER

DPDT signal relay 5 V coil, pair 4-5

+5V >1	COIL_PLUS	COM_A	CUT_4
PHY_L4 >1	NC_A	COM_B	CUT_5
PHY_L5 >1	NC_B	NO_A	T4 >3
RELAY_COIL_LOW	COIL_MINUS	NO_B	T5 >3

K4

BARRIER

DPDT signal relay 5 V coil, pair 7-8

+5V >1	COIL_PLUS	COM_A	CUT_7
PHY_L7 >1	NC_A	COM_B	CUT_8
PHY_L8 >1	NC_B	NO_A	T7 >3
RELAY_COIL_LOW	COIL_MINUS	NO_B	T8 >3

Q1

NPN + 1k base R + flyback diode (coil driver)

+5V >1	FLYBACK_K
GND >1,3	EMITTER
RELAY_COIL_LOW	COLLECTOR
RELAY_DRV >1	BASE_VIA_1k

J2

RJ45 breakout: cable under test

CUT_1	P1
CUT_2	P2
CUT_3	P3
CUT_6	P6
CUT_4	P4
CUT_5	P5
CUT_7	P7
CUT_8	P8

RT1

RT-1 terminator: 4x (1N4148 + R) on an RJ45 plug

CUT_1	P1_D1_ANODE
CUT_2	P2_R_1k0
CUT_3	P3_D2_ANODE
CUT_6	P6_R_2k2
CUT_4	P4_D3_ANODE
CUT_5	P5_R_4k7
CUT_7	P7_D4_ANODE
CUT_8	P8_R_ID

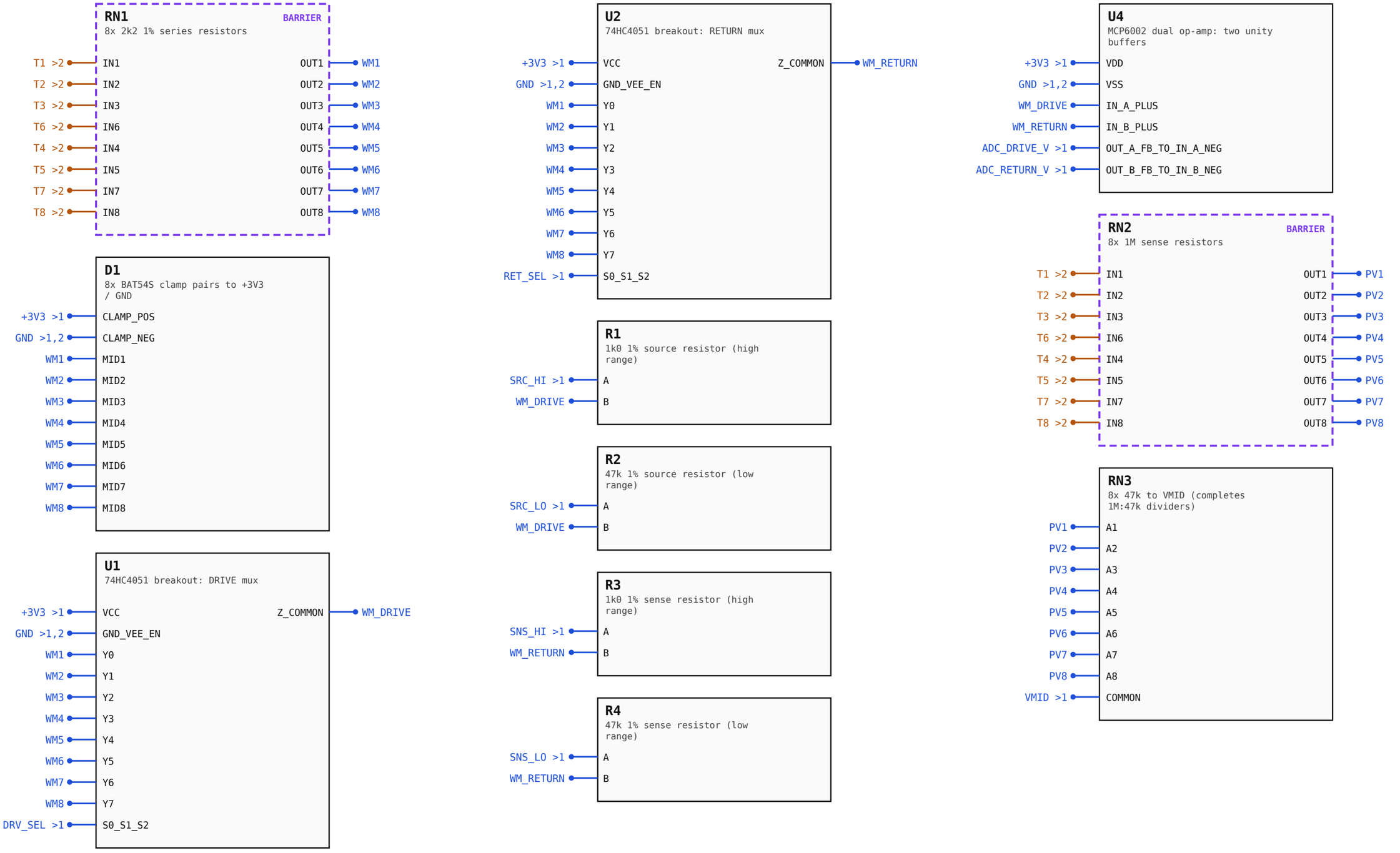
NetScope X1 bench kit - wiring schematic

Relay adapter and cable

Not built. Confirm every eval-board connector against its user guide before power.

SYS system side · LINE cable conductors · POE network-powered

Block level: function-named pins, not package pins



NetScope X1 bench kit - wiring schematic

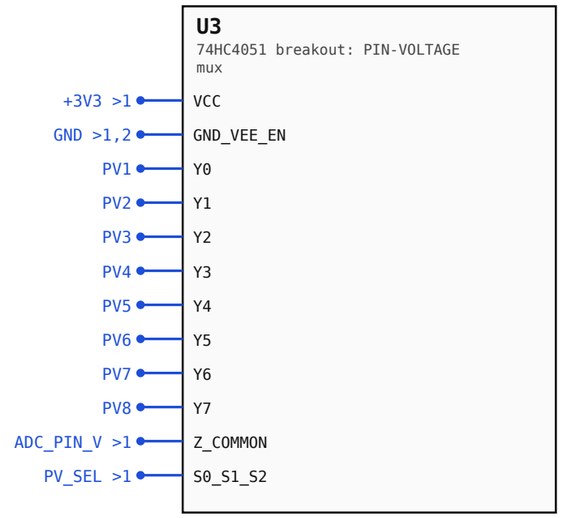
Wiremap front end (breadboard) (1)

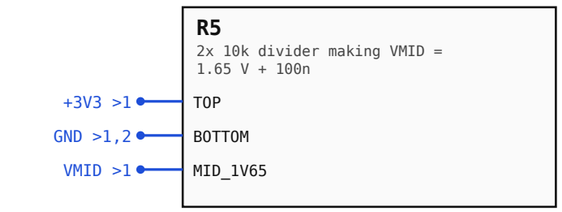
Not built. Confirm every eval-board connector against its user guide before power.

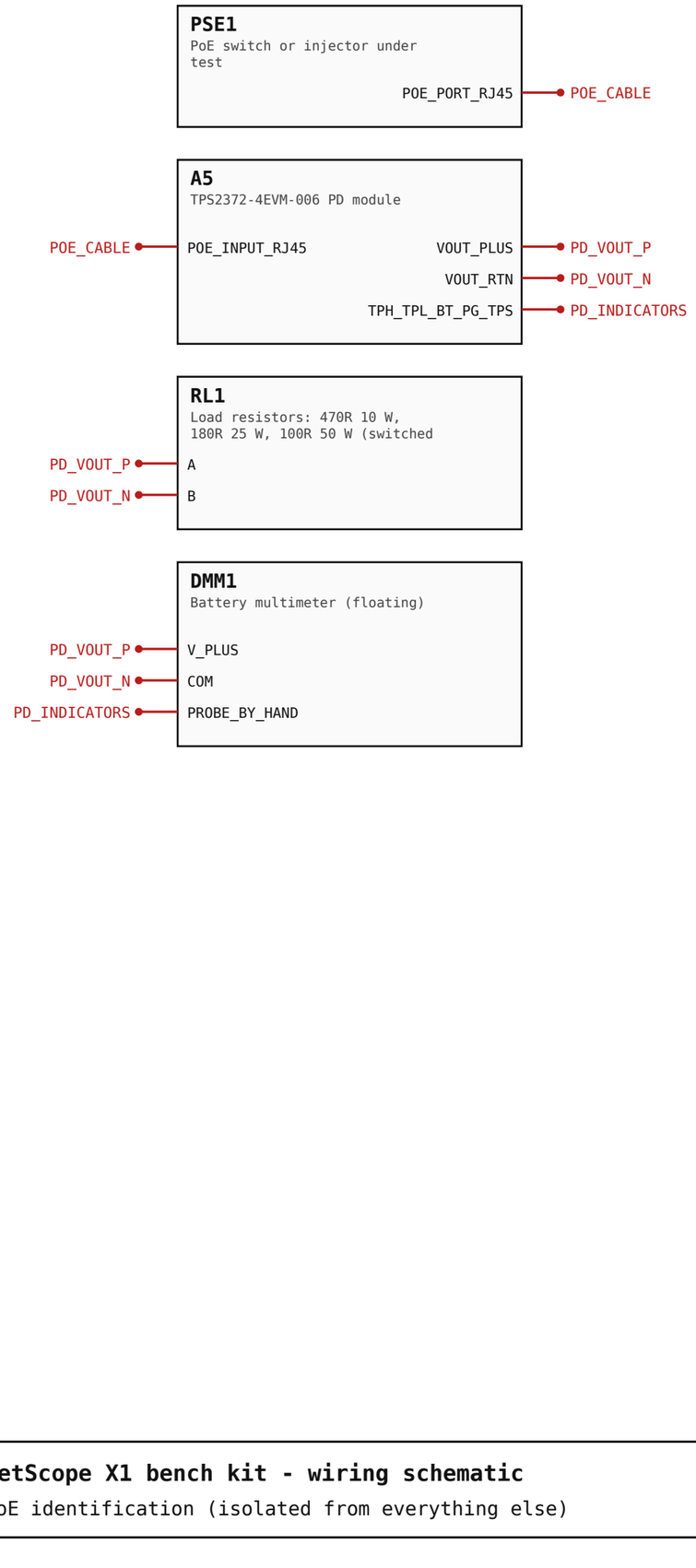
SYS system side · LINE cable conductors · POE network-powered

Sheet 3 of 7

Block level: function-named pins, not package pins


U3
74HC4051 breakout: PIN-VOLTAGE mux
+3V3 >1
VCC
GND >1,2
GND_VEE_EN
PV1
Y0
PV2
Y1
PV3
Y2
PV4
Y3
PV5
Y4
PV6
Y5
PV7
Y6
PV8
Y7
ADC_PIN_V >1
Z_COMMON
PV_SEL >1
S0_S1_S2


R5
2x 10k divider making VMID = 1.65 V + 100n
+3V3 >1
TOP
GND >1,2
BOTTOM
VMID >1
MID_1V65



NetScope X1 bench kit - wiring schematic

PoE identification (isolated from everything else)

Not built. Confirm every eval-board connector against its user guide before power.

SYS system side · LINE cable conductors · POE network-powered

Sheet 5 of 7

Block level: function-named pins, not package pins

Net	Domain	Nodes
USBC_5V1	SYS	PSU1.USB_C, A1.USB_C_POWER
DSI	SYS	A1.DSI1_FFC, A2.DSI_FFC
DISP_5V	SYS	A1.HDR_5V, A2.5V
USB3	SYS	A1.USB3_TYPE_A, A3.USB3_UPSTREAM
EDS_RGMII_MDIO	SYS	A3.EDS_CONNECTOR, A4.EDS_CONNECTOR
COPROC_TX	SYS	A1.GPIO14_TXD, A6.USART_RX
COPROC_RX	SYS	A1.GPIO15_RXD, A6.USART_TX
+5V	SYS	A6.5V, K1.COIL_PLUS, K2.COIL_PLUS, K3.COIL_PLUS, K4.COIL_PLUS, Q1.FLYBACK_K
+3V3	SYS	A6.3V3, U1.VCC, U2.VCC, U3.VCC, U4.VDD, D1.CLAMP_POS, R5.TOP
GND	SYS	A1.HDR_GND, A2.GND, A6.GND, Q1.EMITTER, U1.GND_VEE_EN, U2.GND_VEE_EN, U3.GND_VEE_EN, U4.VSS, D1.CLAMP_NEG, R5.BOTTOM
PHY_L1	LINE	A4.RJ45_P1, J1.P1, K1.NC_A
PHY_L2	LINE	A4.RJ45_P2, J1.P2, K1.NC_B
PHY_L3	LINE	A4.RJ45_P3, J1.P3, K2.NC_A
PHY_L6	LINE	A4.RJ45_P6, J1.P6, K2.NC_B
PHY_L4	LINE	A4.RJ45_P4, J1.P4, K3.NC_A
PHY_L5	LINE	A4.RJ45_P5, J1.P5, K3.NC_B
PHY_L7	LINE	A4.RJ45_P7, J1.P7, K4.NC_A
PHY_L8	LINE	A4.RJ45_P8, J1.P8, K4.NC_B
CUT_1	LINE	K1.COM_A, J2.P1, RT1.P1_D1_ANODE
CUT_2	LINE	K1.COM_B, J2.P2, RT1.P2_R_1k0
CUT_3	LINE	K2.COM_A, J2.P3, RT1.P3_D2_ANODE
CUT_6	LINE	K2.COM_B, J2.P6, RT1.P6_R_2k2
CUT_4	LINE	K3.COM_A, J2.P4, RT1.P4_D3_ANODE
CUT_5	LINE	K3.COM_B, J2.P5, RT1.P5_R_4k7
CUT_7	LINE	K4.COM_A, J2.P7, RT1.P7_D4_ANODE
CUT_8	LINE	K4.COM_B, J2.P8, RT1.P8_R_ID
T1	LINE	K1.NO_A, RN1.IN1, RN2.IN1
T2	LINE	K1.NO_B, RN1.IN2, RN2.IN2
T3	LINE	K2.NO_A, RN1.IN3, RN2.IN3
T6	LINE	K2.NO_B, RN1.IN6, RN2.IN6
T4	LINE	K3.NO_A, RN1.IN4, RN2.IN4
T5	LINE	K3.NO_B, RN1.IN5, RN2.IN5
T7	LINE	K4.NO_A, RN1.IN7, RN2.IN7
T8	LINE	K4.NO_B, RN1.IN8, RN2.IN8
RELAY_COIL_LOW	SYS	Q1.COLLECTOR, K1.COIL_MINUS, K2.COIL_MINUS, K3.COIL_MINUS, K4.COIL_MINUS
RELAY_DRV	SYS	A6.GPIO_RELAY, Q1.BASE_VIA_1k
WM1	SYS	RN1.OUT1, D1.MID1, U1.Y0, U2.Y0
WM2	SYS	RN1.OUT2, D1.MID2, U1.Y1, U2.Y1
WM3	SYS	RN1.OUT3, D1.MID3, U1.Y2, U2.Y2
WM4	SYS	RN1.OUT4, D1.MID4, U1.Y3, U2.Y3
WM5	SYS	RN1.OUT5, D1.MID5, U1.Y4, U2.Y4
WM6	SYS	RN1.OUT6, D1.MID6, U1.Y5, U2.Y5
WM7	SYS	RN1.OUT7, D1.MID7, U1.Y6, U2.Y6
WM8	SYS	RN1.OUT8, D1.MID8, U1.Y7, U2.Y7

Net	Domain	Nodes
SRC_HI	SYS	A6.GPIO_SRC_HI, R1.A
SRC_LO	SYS	A6.GPIO_SRC_LO, R2.A
WM_DRIVE	SYS	R1.B, R2.B, U1.Z_COMMON, U4.IN_A_PLUS
SNS_HI	SYS	A6.GPIO_SNS_HI, R3.A
SNS_LO	SYS	A6.GPIO_SNS_LO, R4.A
WM_RETURN	SYS	R3.B, R4.B, U2.Z_COMMON, U4.IN_B_PLUS
ADC_DRIVE_V	SYS	U4.OUT_A_FB_TO_IN_A_NEG, A6.ADC_IN0
ADC_RETURN_V	SYS	U4.OUT_B_FB_TO_IN_B_NEG, A6.ADC_IN1
DRV_SEL	SYS	A6.GPIO_DRV_S0_S1_S2, U1.S0_S1_S2
RET_SEL	SYS	A6.GPIO_RET_S0_S1_S2, U2.S0_S1_S2
PV1	SYS	RN2.OUT1, RN3.A1, U3.Y0
PV2	SYS	RN2.OUT2, RN3.A2, U3.Y1
PV3	SYS	RN2.OUT3, RN3.A3, U3.Y2
PV4	SYS	RN2.OUT4, RN3.A4, U3.Y3
PV5	SYS	RN2.OUT5, RN3.A5, U3.Y4
PV6	SYS	RN2.OUT6, RN3.A6, U3.Y5
PV7	SYS	RN2.OUT7, RN3.A7, U3.Y6
PV8	SYS	RN2.OUT8, RN3.A8, U3.Y7
VMID	SYS	R5.MID_1V65, RN3.COMMON, A6.ADC_IN3
ADC_PIN_V	SYS	U3.Z_COMMON, A6.ADC_IN2
PV_SEL	SYS	A6.GPIO_PV_S0_S1_S2, U3.S0_S1_S2
POE_CABLE	POE	PSE1.POE_PORT_RJ45, A5.POE_INPUT_RJ45
PD_VOUT_P	POE	A5.VOUT_PLUS, RL1.A, DMM1.V_PLUS
PD_VOUT_N	POE	A5.VOUT_RTN, RL1.B, DMM1.COM
PD_INDICATORS	POE	A5.TPH_TPL_BT_PG_TPS, DMM1.PROBE_BY_HAND